

|           | Q1-2 | Q3-5 | Q6 a | Q6 b – f | Q7 a | Q7 b c | Q7 d |     |
|-----------|------|------|------|----------|------|--------|------|-----|
| <b>E4</b> | /2   |      | /7   |          |      | /9     |      | /18 |
| <b>E8</b> |      | /3   |      | /12      | /3   |        | /5   | 23  |
|           |      |      |      |          |      |        |      | /41 |

## Section I

5 marks

Attempt Questions 1–5

Allow about 7 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5.

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1. Which pair of equations are the directrices of the hyperbola  $16x^2 - 25y^2 = 400$ ?
- (A)  $x = \pm \frac{25}{\sqrt{41}}$
- (B)  $x = \pm \frac{1}{\sqrt{41}}$
- (C)  $x = \pm \frac{\sqrt{41}}{4}$
- (D)  $x = \pm \frac{\sqrt{41}}{25}$
2. What is the equation of the chord of contact from  $(5, -2)$  to the ellipse  $\frac{x^2}{9} + \frac{y^2}{4} = 1$ ?
- (A)  $\frac{x}{2} - \frac{5y}{16} = 1$
- (B)  $\frac{5x}{16} - \frac{2y}{9} = 1$
- (C)  $\frac{5x}{16} + \frac{2y}{9} = 1$
- (D)  $\frac{5x}{9} - \frac{y}{2} = 1$
3. Which expression is equal to  $\int \cot x \, dx$ ?
- (A)  $-\operatorname{cosec}^2 x + C$
- (B)  $-\ln(\operatorname{cosec} x) + C$
- (C)  $\frac{\cot^2 x}{2} + C$
- (D)  $-\ln(\operatorname{cosec} x + \cot x) + C$

4. Given that  $2 \cos A \cos B = \cos(A - B) + \cos(A + B)$ , which of the following is equal to

$$\int \cos 3x \cos 4x \, dx?$$

- (A)  $\frac{1}{2} \sin x + \frac{1}{2} \sin 7x + C$
- (B)  $\frac{1}{2} \cos(-x) + \frac{1}{14} \cos 7x + C$
- (C)  $\frac{1}{2} \sin x + \frac{1}{14} \sin 7x + C$
- (D)  $-\frac{1}{2} \sin(-x) - \frac{1}{14} \sin 7x + C$

5. Which of the following definite integrals is greater than zero for  $0 < a < 1$ ?

- (A)  $\int_{-a}^a \frac{\tan^{-1} x}{\cos x} \, dx$
- (B)  $\int_{-a}^a -\frac{\tan^{-1} x}{\sin^{-1} x} \, dx$
- (C)  $\int_{-a}^a e^{-x^2} \, dx$
- (D)  $\int_{-a}^a \frac{\sin^{-1} x}{1+x^2} \, dx$

**END OF SECTION I**

## Section II

36 marks

Attempt Questions 6 & 7

Allow about 48 minutes for this section

Answer each question in a NEW writing booklet. Extra pages are available

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### Question 6 (19 marks)

- (a) The point  $P(3\sec\theta, \tan\theta)$  lies on an hyperbola with centre at the origin and major axis  $y=0$ .
- (i) Write down the Cartesian equation of the hyperbola. 1
  - (ii) Find the eccentricity of the hyperbola. 1
  - (ii) Draw a neat **ONE-THIRD** page sketch of this hyperbola, showing the coordinates of the vertices and the foci, as well as the equations of the directrices and asymptotes. 3
  - (iv) Derive the equation of the tangent to the hyperbola at the point  $P$  in parametric form. 2
- (b) Find  $\int \cos^3 x \sin^5 x \, dx$  2
- (c) Find  $\int x \tan^{-1} x \, dx$ . 3
- (d) (i) Write  $\frac{3x^2+3x+2}{(x-1)(x^2+1)}$  in the form  $\frac{A}{x-1} + \frac{Bx+C}{x^2+1}$  2
- (ii) Hence find  $\int \frac{3x^2+3x+2}{(x-1)(x^2+1)} \, dx$  3
- (e) Find  $\int \frac{dx}{\sqrt{5-8x-4x^2}}$  2

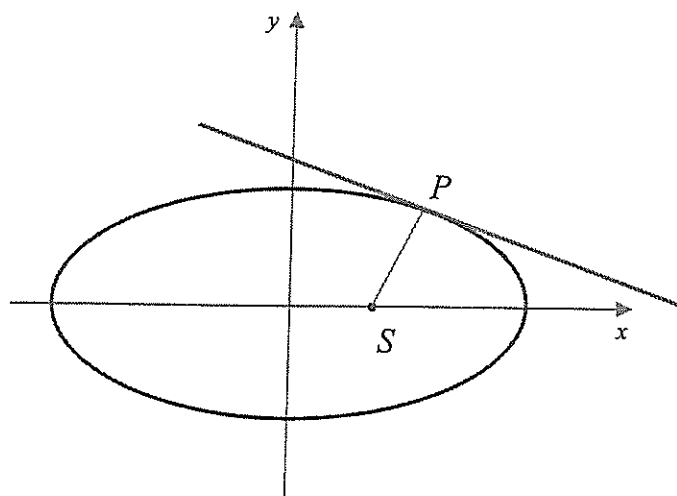
End of Question 6

**Question 7** (17 marks)

(a) Evaluate  $\int_0^{\frac{\pi}{2}} \frac{d\theta}{\cos \theta + 2}$  using the substitution  $t = \tan \frac{\theta}{2}$

3

- (b) The diagram below shows the variable point  $P(a \cos \theta, b \sin \theta)$  on the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  and the focus  $S$  on the positive  $x$ -axis.



- (i) Find the gradient of the tangent at  $P$  in terms of  $a$ ,  $b$  and  $\theta$ .

1

- (ii) Show that the product of the gradient of  $SP$  and the gradient of the tangent at  $P$  is:

2

$$\frac{\cos \theta (1 - e^2)}{e - \cos \theta}$$

- (iii) If  $P$  is at the end points of the major axis then  $PS$  is perpendicular to the tangent at  $P$ .

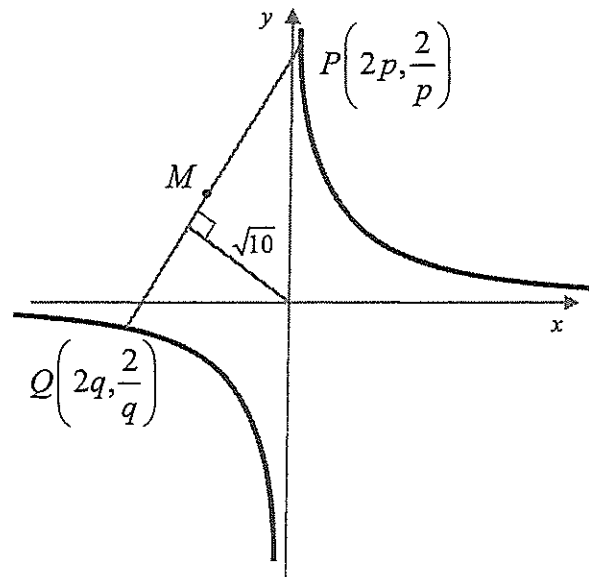
2

Use part (ii) to show that there are no other points on the curve where  $PS$  is perpendicular to the tangent at  $P$ .

**Question 7 continues on the next page.**

**Question 7** (Continued)

- (c) In the diagram below  $P\left(2p, \frac{2}{p}\right)$  and  $Q\left(2q, \frac{2}{q}\right)$  are variable points on the rectangular hyperbola  $xy = 4$ .  $M$  is the midpoint of the chord  $PQ$  and the perpendicular distance from  $PQ$  to the origin is fixed at  $\sqrt{10}$  units.



Given that  $PQ$  has equation  $x + pqr = 2(p + q)$

- (i) Show that  $4(p + q)^2 = 10(1 + p^2q^2)$  2
- (ii) Find the Cartesian equation of the locus of  $M$ . 2  
You are not required to consider the restrictions on this locus.

(d) Let  $I_n = \int_1^2 \left(1 - \frac{1}{x}\right)^n dx$  for all positive integers  $n$ .

(i) Show that  $I_{n+1} = \frac{n+1}{n} I_n - \frac{1}{n \times 2^n}$  3

(ii) It can be shown that:

$$\frac{1}{n+1} I_{n+1} = I_1 - \sum_{r=1}^n \frac{1}{r(r+1)2^r} \quad (\text{DO NOT PROVE THIS})$$

Hence find the limiting sum of the series: 2

$$\frac{1}{1 \times 2 \times 2^1} + \frac{1}{2 \times 3 \times 2^2} + \frac{1}{3 \times 4 \times 2^3} + \frac{1}{4 \times 5 \times 2^4} + \dots$$

**END OF EXAMINATION**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x$ ,  $x > 0$

## Section I

1. Which pair of equations are the directrices of the hyperbola  $16x^2 - 25y^2 = 400$ ?

(A)  $x = \pm \frac{25}{\sqrt{41}}$

(B)  $x = \pm \frac{1}{\sqrt{41}}$

(C)  $x = \pm \frac{\sqrt{41}}{4}$

(D)  $x = \pm \frac{\sqrt{41}}{25}$

|                                                                                                                                                                                                      |                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $16x^2 - 25y^2 = 400$<br>$\frac{x^2}{25} - \frac{y^2}{16} = 1$<br>$a^2 = 25$ and $b^2 = 16$<br>$16 = 25(e^2 - 1)$<br>$\frac{16}{25} = e^2 - 1$<br>$\frac{41}{25} = e^2$<br>$e = \frac{\sqrt{41}}{5}$ | Directrices are:<br>$x = \pm \frac{a}{e}$<br>$= \pm \frac{5}{\frac{\sqrt{41}}{5}}$<br>$= \pm \frac{25}{\sqrt{41}}$ |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|

2. What is the equation of the chord of contact from  $(5, -2)$  to the ellipse  $\frac{x^2}{9} + \frac{y^2}{4} = 1$ ?

(A)  $\frac{x}{2} - \frac{5y}{16} = 1$

(B)  $\frac{5x}{16} - \frac{2y}{9} = 1$

(C)  $\frac{5x}{16} + \frac{2y}{9} = 1$

(D)  $\frac{5x}{9} - \frac{y}{2} = 1$

|                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------|
| Chord of contact is<br>$\frac{x_0x}{9} + \frac{y_0y}{4} = 1$<br>$\frac{5x}{9} - \frac{2y}{4} = 1$<br>$\frac{5x}{9} - \frac{y}{2} = 1$ |
|---------------------------------------------------------------------------------------------------------------------------------------|

3. Which expression is equal to  $\int \cot x \, dx$ ?

(A)  $-\operatorname{cosec}^2 x + C$

(B)  $-\ln(\operatorname{cosec} x) + C$

(C)  $\frac{\cot^2 x}{2} + C$

(D)  $-\ln(\operatorname{cosec} x + \cot x) + C$

$$\begin{aligned}\int \cot x \, dx &= \int \frac{\cos x}{\sin x} \, dx \\ &= \ln(\sin x) + C \\ &= \ln\left(\frac{1}{\sin x}\right)^{-1} + C \\ &= \ln(\operatorname{cosec} x)^{-1} + C \\ &= -\ln(\operatorname{cosec} x) + C\end{aligned}$$

4. Given that  $2 \cos A \cos B = \cos(A - B) + \cos(A + B)$ , which of the following is equal to

$$\int \cos 3x \cos 4x \, dx?$$

(A)  $\frac{1}{2} \sin x + \frac{1}{2} \sin 7x + C$

(B)  $\frac{1}{2} \cos(-x) + \frac{1}{14} \cos 7x + C$

(C)  $\frac{1}{2} \sin x + \frac{1}{14} \sin 7x + C$

(D)  $-\frac{1}{2} \sin(-x) - \frac{1}{14} \sin 7x + C$

$$\begin{aligned}\int \cos 3x \cos 4x \, dx &= \frac{1}{2} \int \cos(4x - 3x) + \cos(4x + 3x) \\ &= \frac{1}{2} \int \cos(x) + \cos(7x) \\ &= \frac{1}{2} \left( \sin(x) + \frac{\sin(7x)}{7} \right) + C \\ &= \frac{1}{2} \sin x + \frac{1}{14} \sin 7x + C\end{aligned}$$

5. Which of the following definite integrals is greater than zero for  $0 < a < 1$ ?

(A)  $\int_{-a}^a \frac{\tan^{-1} x}{\cos x} \, dx$

(B)  $\int_{-a}^a -\frac{\tan^{-1} x}{\sin^{-1} x} \, dx$

(C)  $\int_{-a}^a e^{-x^2} \, dx$

(D)  $\int_{-a}^a \frac{\sin^{-1} x}{1 + x^2} \, dx$

Since  $\frac{\tan^{-1} x}{\cos x}$ ,  $-\frac{\tan^{-1} x}{\sin^{-1} x}$  and  $\frac{\sin^{-1} x}{1 + x^2}$  are all odd functions then the integrals for these will all be zero.

And since  $e^{-x^2}$  is an even function and always positive then  $\int_{-a}^a e^{-x^2} \, dx > 0$

**Question 6** (19 marks)

- (a) The point  $P(3\sec\theta, \tan\theta)$  lies on an hyperbola with centre at the origin and major axis  $y = 0$ .

- (i) Write down the Cartesian equation of the hyperbola.

1

$$\frac{(3\sec\theta)^2}{9} - (\tan\theta)^2 = 1$$

$$\frac{x^2}{9} - y^2 = 1$$

- (ii) Find the eccentricity of the hyperbola.

1

$$\frac{x^2}{9} - y^2 = 1$$

$$a^2 = 9 \text{ and } b^2 = 1$$

$$1 = 9(e^2 - 1)$$

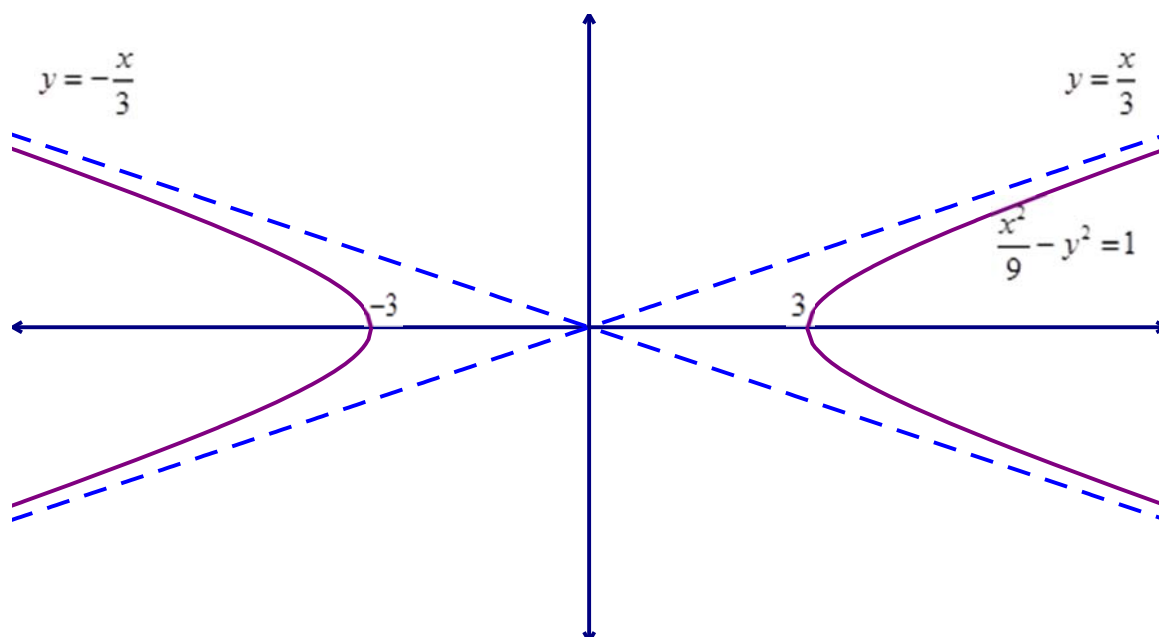
$$\frac{1}{9} = e^2 - 1$$

$$\frac{10}{9} = e^2$$

$$e = \frac{\sqrt{10}}{3} \quad \text{since } e > 0$$

- (ii) Draw a neat **ONE-THIRD** page sketch of this hyperbola, showing the coordinates of the vertices and the foci, as well as the equations of the directrices and asymptotes.

3



(iv) Derive the equation of the tangent to the hyperbola at the point  $P$  in parametric form.

2

|                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>By differentiating parametrically</p> $x = 3 \sec \theta$ $\frac{dx}{d\theta} = 3 \sec \theta \tan \theta$ $y = \tan \theta$ $\frac{dy}{d\theta} = \sec^2 \theta$ $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$ $= \frac{\sec^2 \theta}{3 \sec \theta \tan \theta}$ $= \frac{\sec \theta}{3 \tan \theta}$ | <p>The equation of the tangent is :</p> $y - \tan \theta = \frac{\sec \theta}{3 \tan \theta} (x - 3 \sec \theta)$ $3y \tan \theta - 3 \tan^2 \theta = x \sec \theta - 3 \sec^2 \theta$ $3 \sec^2 \theta - 3 \tan^2 \theta = x \sec \theta - 3y \tan \theta$ $3(\sec^2 \theta - \tan^2 \theta) = x \sec \theta - 3y \tan \theta$ $1 = \frac{x \sec \theta}{3} - y \tan \theta$ |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

(b) Find  $\int \cos^3 x \sin^5 x dx$

2

$$\begin{aligned}
 \int \cos^3 x \sin^5 x dx &= \int \cos x \cos^2 x \sin^5 x dx \\
 &= \int \cos x (1 - \sin^2 x) \sin^5 x dx \\
 &= \int \cos x \sin^5 x dx - \int \cos x \sin^7 x dx \\
 &= \frac{\sin^6 x}{6} - \frac{\sin^8 x}{8} + C
 \end{aligned}$$

(c) Find  $\int x \tan^{-1} x dx$

3

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $  \begin{aligned}  \int x \tan^{-1} x dx &= \int \frac{d}{dx} \left( \frac{1}{2} x^2 \right) \tan^{-1} x dx \\  &= \frac{1}{2} x^2 \tan^{-1} x - \int \left( \frac{1}{2} x^2 \right) \left( \frac{1}{1+x^2} \right) dx \\  &= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} dx \\  &= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} \int \frac{x^2+1-1}{1+x^2} dx \\  &= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} \int 1 - \frac{1}{1+x^2} dx \\  &= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} x + \frac{1}{2} \tan^{-1} x dx + C  \end{aligned}  $ |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

(d) (i) Write  $\frac{3x^2+3x+2}{(x-1)(x^2+1)}$  in the form  $\frac{A}{x-1} + \frac{Bx+C}{x^2+1}$

2

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $\frac{3x^2+3x+2}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$ $= \frac{A(x^2+1) + (Bx+C)(x-1)}{(x-1)(x^2+1)}$ $3x^2+3x+2 = A(x^2+1) + (Bx+C)(x-1)$ <p>Let <math>x=1</math></p> $3+3+2 = A(1+1) + 0$ $8 = 2A$ $A = 4$ <p>Let <math>x=0</math></p> $2 = A(0+1) + (0+C)(0-1)$ $2 = A - C$ $2 = 4 - C$ $C = 2$ <p>Let <math>x=-1</math></p> $3-3+2 = A(1+1) + (-B+C)(-2)$ $2 = 2A + 2B - 2C$ $2 = 8 + 2B - 4$ $-2 = 2B$ $B = -1$ $\frac{3x^2+3x+2}{(x-1)(x^2+1)} = \frac{8}{x-1} + \frac{-x+2}{x^2+1}$ |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

(ii) Hence find  $\int \frac{3x^2+3x+2}{(x-1)(x^2+1)} dx$

3

|                                                                                                                                                                                                                                                                                                                                                                                                      |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $\int \frac{3x^2+3x+2}{(x-1)(x^2+1)} dx = \int \left( \frac{8}{x-1} + \frac{-x+2}{x^2+1} \right) dx$ $= \int \frac{8}{x-1} dx + \int \frac{-x+2}{x^2+1} dx$ $= \int \frac{8}{x-1} dx - \int \frac{x}{x^2+1} dx + \int \frac{2}{x^2+1} dx$ $= 8 \int \frac{1}{x-1} dx - \frac{1}{2} \int \frac{2x}{x^2+1} dx + 2 \int \frac{1}{x^2+1} dx$ $= 8 \ln(x-1) - \frac{1}{2} \ln(x^2+1) + 2 \tan^{-1} x + C$ |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

(e) Find  $\int \frac{dx}{\sqrt{5-8x-4x^2}}$

2

|                                                                                                                                                                                                                                                                    |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $\begin{aligned}\int \frac{dx}{\sqrt{5-8x-4x^2}} &= \int \frac{dx}{\sqrt{9-4(x^2+2x+1)}} \\ &= \int \frac{dx}{\sqrt{9-4(x+1)^2}} \\ &= \int \frac{dx}{2\sqrt{\left(\frac{3}{2}\right)^2 - (x+1)^2}} \\ &= \frac{1}{2} \sin^{-1} \frac{2(x+1)}{3} + C\end{aligned}$ |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

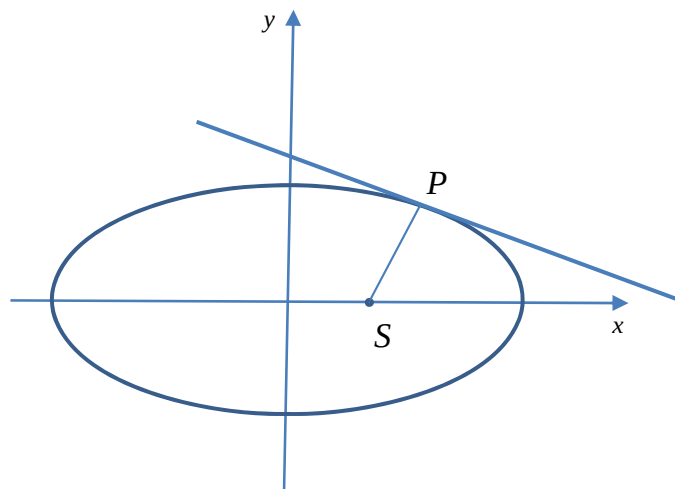
**Question 7** (17 marks)

(a) Evaluate  $\int_0^{\frac{\pi}{2}} \frac{d\theta}{\cos\theta+2}$  using the substitution  $t = \tan \frac{\theta}{2}$

3

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\begin{aligned}\int_0^{\frac{\pi}{2}} \frac{d\theta}{\cos\theta+2} \\ \int_0^1 \frac{1}{\frac{1-t^2}{1+t^2}+2} \times \frac{2}{1+t^2} dt &= \int_0^1 \frac{2}{1-t^2+2(1+t^2)} dt \\ &= \int_0^1 \frac{2}{3+t^2} dt \\ &= \left[ \frac{2}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1 \\ &= \frac{2}{\sqrt{3}} \left[ \tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} \frac{0}{\sqrt{3}} \right] \\ &= \frac{2}{\sqrt{3}} \times \frac{\pi}{6} = \frac{\pi}{3\sqrt{3}}\end{aligned}$ | $\begin{aligned}t &= \tan \frac{\theta}{2} \\ \frac{\theta}{2} &= \tan^{-1} t \\ \theta &= 2 \tan^{-1} t \\ \frac{d\theta}{dt} &= \frac{2}{1+t^2} \\ d\theta &= \frac{2dt}{1+t^2}\end{aligned}$ <p>When <math>\theta = \frac{\pi}{2}</math>      <math>t = \tan \frac{\pi}{4} = 1</math><br/> <math>\theta = 0</math>              <math>t = \tan 0 = 0</math></p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- (b) The diagram below shows the variable point  $P(a \cos \theta, b \sin \theta)$  on the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  and the focus  $S$  on the positive  $x$ -axis.



- (i) Find the gradient of the tangent at  $P$  in terms of  $a$ ,  $b$  and  $\theta$ .

1

|                                                                                                                                                                                                                   |                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ <p>Differentiate Implicitly</p> $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$ $\frac{y}{b^2} \frac{dy}{dx} = -\frac{x}{a^2}$ $\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$ | <p>At the Point <math>P(a \cos \theta, b \sin \theta)</math></p> $\frac{dy}{dx} = -\frac{b^2 a \cos \theta}{a^2 b \sin \theta}$ $m_{Tangent} = -\frac{b \cos \theta}{a \sin \theta}$ |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- (ii) Show that the product of the gradient of  $SP$  and the gradient of the tangent at  $P$  is:

2

$$\frac{\cos \theta (1 - e^2)}{e - \cos \theta}$$

|                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Gradient of <math>PS</math> <math>P(a \cos \theta, b \sin \theta)</math> and <math>S(ae, 0)</math></p> $m_{PS} = \frac{b \sin \theta - 0}{a \cos \theta - ae}$ $= \frac{b \sin \theta}{a \cos \theta - ae}$ | $m_{PS} \times m_{Tangent} = \frac{b \sin \theta}{a \cos \theta - ae} \times -\frac{b \cos \theta}{a \sin \theta}$ $= -\frac{b^2 \cos \theta}{a^2 (\cos \theta - e)}$ $= \frac{a^2 (1 - e^2) \cos \theta}{a^2 (e - \cos \theta)}$ $= \frac{(1 - e^2) \cos \theta}{(e - \cos \theta)}$ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

(iii) If  $P$  is at the end points of the major axis then  $PS$  is perpendicular to the tangent at  $P$ .

2

Use part (ii) to show that there are no other points on the curve where  $PS$  is perpendicular to the tangent at  $P$ .

Let  $m_{PS} \times m_{Tangent} = -1$

$$\frac{\cos \theta (1 - e^2)}{e - \cos \theta} = -1$$

$$\cos \theta (1 - e^2) = \cos \theta - e$$

$$-e^2 \cos \theta = -e$$

$$\cos \theta = \frac{e}{e^2}$$

$$\cos \theta = \frac{1}{e}$$

Since  $0 < e < 1$  For an ellipse

Then  $\frac{1}{e} > 1$

Therefore there are no values for  $\theta$  where  $\cos \theta > 1$

letting the product of the two gradients be negative 1 is only valid when neither the tangent at  $P$  nor  $PS$  is vertical.

When  $PS$  is vertical then

$$a \cos \theta = ae$$

$$\cos \theta = e$$

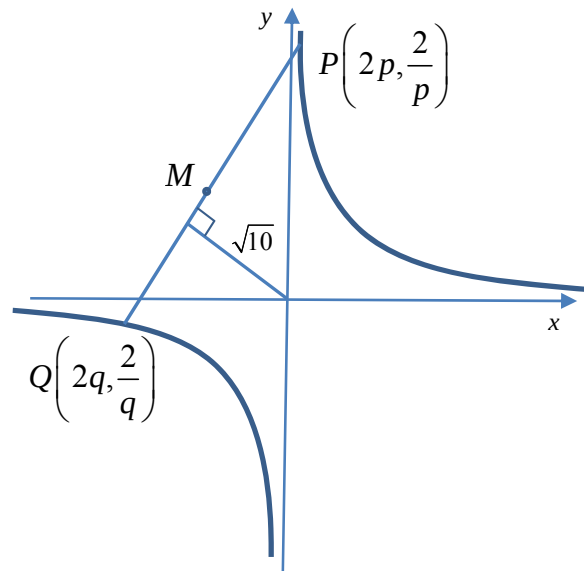
Thus the gradient of the tangent  $m_{Tangent}$  is not zero, thus the tangent and  $PS$  are not perpendicular.

Thus the only place where the tangent and  $PS$  are perpendicular is at the end points of the major axis

**Question 7 continues on the next page.**

**Question 7** (Continued)

- (c) In the diagram below  $P\left(2p, \frac{2}{p}\right)$  and  $Q\left(2q, \frac{2}{q}\right)$  are variable points on the rectangular hyperbola  $xy = 4$ .  $M$  is the midpoint of the chord  $PQ$  and the perpendicular distance from  $PQ$  to the origin is fixed at  $\sqrt{10}$  units.



Given that  $PQ$  has equation  $x + pqy = 2(p + q)$

- (i) Show that  $4(p + q)^2 = 10(1 + p^2q^2)$

2

Equation of  $PQ$  is  $x + pqy = 2(p + q)$   
Then perpendicular distance from  $PQ$  to  $O$

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$\sqrt{10} = \frac{|(1)(0) + (pq)(0) - 2(p + q)|}{\sqrt{(1)^2 + (pq)^2}}$$

$$10 = \frac{4(p + q)^2}{1 + (pq)^2}$$

$$4(p + q)^2 = 10(1 + p^2q^2)$$

- (ii) Find the Cartesian equation of the locus of  $M$ .  
You are not required to consider the restrictions on this locus.

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|                                                                                                                                                                                     |                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The Midpoint <math>M</math> of <math>PQ</math>:</p> $x = \frac{2p+2q}{2} = p+q$ $y = \frac{\frac{2}{p} + \frac{2}{q}}{2} = \frac{p+q}{pq}$ $y = \frac{x}{pq}$ $pq = \frac{x}{y}$ | $4(p+q)^2 = 10(1+p^2q^2) \text{ From part i)}$ $4x^2 = 10 + \frac{10x^2}{y^2}$ $4x^2y^2 = 10y^2 + 10x^2$ $4x^2y^2 - 10y^2 = 10x^2$ $y^2 = \frac{10x^2}{4x^2 - 10}$ $y^2 = \frac{5x^2}{2x^2 - 5}$ |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- (d) Let  $I_n = \int_1^2 \left(1 - \frac{1}{x}\right)^n dx$  for all positive integers  $n$ .

- (i) Show that  $I_{n+1} = \frac{n+1}{n} I_n - \frac{1}{n \times 2^n}$

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### Method 1

$$\begin{aligned}
 I_n &= \int_1^2 \left(1 - \frac{1}{x}\right)^n dx = \int_1^2 \left(\frac{d}{dx}(x)\right) \left(1 - \frac{1}{x}\right)^n dx \\
 &= \left[ x \left(1 - \frac{1}{x}\right)^n \right]_1^2 - \int_1^2 x n \left(1 - \frac{1}{x}\right)^{n-1} \left(-\frac{1}{x^2}\right) dx \\
 &= 2 \left(1 - \frac{1}{2}\right)^n - 1 \left(1 - \frac{1}{1}\right)^n - n \int_1^2 \frac{1}{x} \left(1 - \frac{1}{x}\right)^{n-1} dx \\
 &= \left(\frac{2}{2^n}\right) - (0) - n \int_1^2 \frac{1}{x} \left(1 - \frac{1}{x}\right)^{n-1} dx \\
 &= \left(\frac{1}{2^{n-1}}\right) + n \int_1^2 \left(1 - \frac{1}{x} - 1\right) \left(1 - \frac{1}{x}\right)^{n-1} dx \\
 &= \left(\frac{1}{2^{n-1}}\right) + n \int_1^2 \left(1 - \frac{1}{x}\right)^n - \left(1 - \frac{1}{x}\right)^{n-1} dx \\
 I_n &= \left(\frac{1}{2^{n-1}}\right) + nI_n - nI_{n-1} \\
 I_n - nI_n &= \frac{1}{2^{n-1}} - nI_{n-1} \\
 I_n(n-1) &= nI_{n-1} - \frac{1}{2^{n-1}}
 \end{aligned}$$

Replace  $n$  with  $n-1$

$$I_{n+1}(n+1-1) = (n+1)I_{n+1-1} - \frac{1}{2^{n+1-1}}$$

$$nI_{n+1} = (n+1)I_n - \frac{1}{2^n}$$

$$I_{n+1} = \frac{(n+1)}{n}I_n - \frac{1}{n2^n}$$

## Method 2

$$\begin{aligned} I_n &= \int_1^2 \left(1 - \frac{1}{x}\right)^n dx = \int_1^2 \left(\frac{x-1}{x}\right)^n dx \\ &= \int_1^2 \frac{(x-1)^n}{x^n} dx \\ &= \int_1^2 \frac{d}{dx} \left( \frac{(x-1)^{n+1}}{n+1} \right) \left( \frac{1}{x^n} \right) dx \\ &= \int_1^2 \frac{d}{dx} \left( \frac{(x-1)^{n+1}}{n+1} \right) (x^{-n}) dx \\ &= \left[ \frac{(x-1)^{n+1}}{n+1} \frac{1}{x^n} \right]_1^2 - \int_1^2 \frac{(x-1)^{n+1}}{n+1} (-nx^{-n-1}) dx \\ &= \left[ \frac{(x-1)^{n+1}}{n+1} \frac{1}{x^n} \right]_1^2 + \frac{n}{n+1} \int_1^2 \frac{(x-1)^{n+1}}{x^{n+1}} dx \\ &= \left[ \left( \frac{(2-1)^{n+1}}{n+1} \frac{1}{2^n} \right) - (0) \right] + \frac{n}{n+1} I_{n+1} \end{aligned}$$

$$I_n = \frac{1}{(n+1)2^n} + \frac{n}{n+1} I_{n+1}$$

$$(n+1)I_n = \frac{1}{2^n} + nI_{n+1}$$

$$\left( \frac{n+1}{n} \right) I_n = \frac{1}{n2^n} + I_{n+1}$$

$$I_{n+1} = \left( \frac{n+1}{n} \right) I_n - \frac{1}{n2^n}$$

(ii) It can be shown that:

$$\frac{1}{n+1}I_{n+1} = I_1 - \sum_{r=1}^n \frac{1}{r(r+1)2^r} \quad (\text{DO NOT PROVE THIS})$$

Hence find the limiting sum of the series:

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$$\frac{1}{1 \times 2 \times 2^1} + \frac{1}{2 \times 3 \times 2^2} + \frac{1}{3 \times 4 \times 2^3} + \frac{1}{4 \times 5 \times 2^4} + \dots$$

Solution

$$\frac{1}{1 \times 2 \times 2^1} + \frac{1}{2 \times 3 \times 2^2} + \frac{1}{3 \times 4 \times 2^3} + \frac{1}{4 \times 5 \times 2^4} + \dots = \sum_{r=1}^{\infty} \frac{1}{r(r+1)2^r}$$

$$\frac{1}{n+1}I_{n+1} = I_1 - \sum_{r=1}^n \frac{1}{r(r+1)2^r}$$

$$\sum_{r=1}^n \frac{1}{r(r+1)2^r} = I_1 - \frac{1}{n+1}I_{n+1}$$

As  $n \rightarrow \infty$

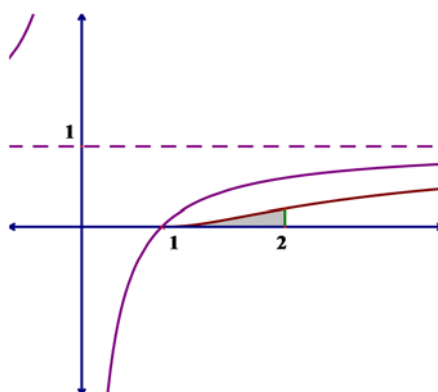
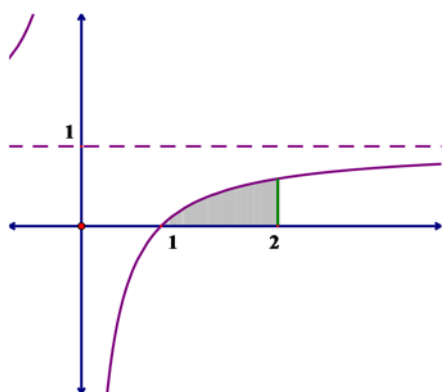
$$\frac{1}{n+1} \rightarrow 0$$

From the Graph below:

For  $1 < x < 2$   $0 < 1 - \frac{1}{x} < 1$

$$\therefore 0 < \left(1 - \frac{1}{x}\right)^n < 1 - \frac{1}{x}$$

$$\therefore I_{n+1} < I_n \text{ for all } n > 1$$



As  $n \rightarrow \infty$

$$I_{n+1} \rightarrow 0$$

$$\frac{1}{1 \times 2 \times 2^1} + \frac{1}{2 \times 3 \times 2^2} + \frac{1}{3 \times 4 \times 2^3} + \frac{1}{4 \times 5 \times 2^4} + \dots = I_1$$

$$= \int_1^2 \left(1 - \frac{1}{x}\right)^n dx$$

$$= [x - \ln x]_1^2$$

$$= 1 - \ln 2$$